

THE PROBLEM, IN INCOIS'S OWN WORDS

INCOIS produces a three dimensional ocean every single day: temperature, salinity, currents and chlorophyll as model output, plus live instruments in the water (Argo floats, gliders, CTD casts, BGC floats). Their own problem statement says:

- "No integrated, web-based 3D visualization platform currently exists that can simultaneously render model fields and in-situ instrument observations in a single interactive environment."
- Existing tools are "desktop-bound, support only 2D plan views, or lack the ability to co-visualize model outputs alongside instrument profiles."
- Forecasters are "forced to toggle between disparate software packages," which "impedes timely hazard assessment, search-and-rescue support, fishery advisories, climate monitoring."
- Never say "INCOIS has no 3D".** Their Digital Ocean portal advertises 3D and 4D visualisation, and one informed judge sinks the pitch on that sentence. The defensible claim is the PS's own: the model and the instruments do not live in one place, so a forecaster has to change software to compare them.

THE FIVE THINGS WE ARE SOLVING

The PS names five gap areas. These are our five points, in this order.

#	The gap, in the PS's terms	What we build for it
1	No web-based 3D volumetric rendering of the water column	The Bay of Bengal as 24 stacked depth levels, 5 m to 2000 m, in a browser, plus isosurface extraction and time animation
2	No unified display of instruments alongside model fields	Every Argo, BGC and moored-buoy cast on the same globe, clickable into a depth profile against the model at the same place and time
3	No interactive variable, depth, time and colorbar controls	Variable selector, depth cursor over all 24 levels, time scrubber, editable colour scale (palette, min/max, log or linear), vertical exaggeration
4	No ingestion of new data streams without re-engineering	A YAML registry plus a plugin interface: a new source or a new derived field is a config entry and a file, not a rewrite
5	No tools for rapid, intuitive understanding of 3D ocean phenomena	Guided tours that drive the scene, an assistant that answers only from the data and cites it, and a model-versus-instrument skill number on the same screen as the field

WHERE WE ACTUALLY ARE

The PS lists seven explicit requirements. A "partly met" names the missing clause rather than hiding it.

#	Requirement	Verdict	The gap, if any
F1	3D volumetric rendering	Partly met	All three named techniques built: depth slices, isosurface extraction, time animation. Chlorophyll renders as instrument data but not as a gridded field, for a reason we can defend
F2	Instrument data overlay	Met	All four classes the PS names are live as distinct symbols: Argo floats, BGC floats, a glider and shipboard CTD casts, plus a moored buoy. The glider and the CTD casts are older than the model field, so they are ringed, dated and refused by the verification rather than left to pass for current
F3	Multi-format data ingestion	Met	None
F4	Customisable colorbar and variable controls	Met	None
F5	Web-based, scalable architecture	Partly met	Docker files written, never executed, because Docker is not installed on the build laptop
F6	Extensible design	Met	None. Two live proofs: a moored-buoy reader, and two derived fields computed by plugins
F7	Open standards (OGC WMS / WCS, CF)	Met	None

Four of seven have no gap. Three have a named gap, and two of those three need data or a machine rather than code.

What that is built on

567

automated tests: 485 data plane, 71 assistant, 11 real browser

51 fps

median at 1920x1080 on integrated Intel graphics, not a gaming GPU

27 fps

worst single frame in five seconds, against a 30 fps floor

0

requests leave the machine; a test fails the build if one does

THE PARAMETERS: 13 SERVED TODAY

Six gridded fields, the 3D model side

Name	What it is	Units	Source	Stored or computed
TEMP	Sea water temperature	degC	INCOIS	stored
SAL	Practical salinity	PSU	INCOIS	stored
SIG0	Potential density anomaly (sigma-0)	kg/m3	from TEMP + SAL	plugin , TEOS-10
D26	Depth of the 26 degC isotherm	m	from TEMP	plugin
uo	Eastward current velocity	m/s	Copernicus GLORYS	stored
vo	Northward current velocity	m/s	Copernicus GLORYS	stored

Seven measured instrument parameters, the in-situ side

Name	What it is	Carried by	QC-passed levels
temp	Temperature at each level of a cast	all instruments	12,884
psal	Practical salinity at each level	Argo and BGC floats	12,849
doxy	Dissolved oxygen	BGC floats only	3,142
chla	Chlorophyll-a, a proxy for phytoplankton	BGC floats only	2,120
nitrate	Dissolved nitrate, a nutrient	BGC floats only	327
bbp700	Particle backscatter at 700 nm	BGC floats only	1,955
ph	Acidity of the sea water	BGC floats only	1,957

Plus two vertical coordinates on every cast: **pres** (pressure in decibars, what the instrument actually measures) and **depth** (metres, computed from pressure and latitude by TEOS-10). Say **thirteen** if asked, then offer the split. Do not say twenty.

The three you will be asked to explain

Temperature. How warm the water is at every depth. Bay of Bengal surface in July is about 29 degC; at 2000 m it is about 2.6 degC. It matters because warm water is the fuel a cyclone runs on.

Salinity. How much dissolved salt, on the Practical Salinity Scale. Open ocean is about 35; the Bay of Bengal surface is about 32 to 33 because the Ganges, Brahmaputra and Irrawaddy pour fresh water into it during the monsoon. It is the freshest large bay in the world, and that one fact explains most of its behaviour. It is formally dimensionless, which is why the data file declares its unit as "1" and we print "PSU" on screen.

Density (sigma-0). How heavy a given volume of sea water is. It is what actually stratifies the ocean: heavy sinks, light floats, and the arrangement decides whether a column mixes or stays in layers. Four things to be able to say: it is **never measured** (there is no density sensor, it is computed from temperature, salinity and pressure); we use **TEOS-10**, the international standard since 2009; **sigma-0** means referenced to the surface with 1000 subtracted, which is why our numbers read 20 to 28 and not 1020 to 1028; in our cube it runs 19.8 at the surface to 27.8 at 2000 m.

THE DATASETS, AND HOW MUCH DATA

Source	What it is	Status
INCOIS ERDDAP	10-day gridded Argo analysis, Variational Analysis Method. INCOIS's own product	Live, open
Argo GDAC (Ifremer)	Daily Indian Ocean profile files	Live, open
Argo BGC (Sprof)	Biogeochemical float synthetic profiles	Live, open
RAMA moored buoys	The 90 E line in the Bay of Bengal, which India helps run	Live, open
Copernicus Marine GLORYS	Currents at one twelfth of a degree, 40 levels	Live, free account
NDMA SACHET	India's national CAP disaster-alert backbone, 99 live alerts when ingested	Live, open
INCOIS Oceansat-2 ocean colour	Gridded chlorophyll	Registered and deliberately disabled

Quantity	Number
INCOIS analysis upstream: 813 times x 24 depths x 60 lats x 90 lons	105,321,600 values per variable
Our demo cube, INCOIS temperature and salinity	24,192 values each
Our demo cube, Copernicus eastward and northward current	5,234,520 values each
Total gridded values on disk	10,517,424
Instrument casts, and the instruments that took them	149 casts from 19 instruments (13 Argo floats, 3 BGC floats casting 3 times each, 1 RAMA buoy casting 3 times, 1 glider diving 109 times, 1 research vessel casting 15 times)
Measurement levels that passed quality control	30,012, filtered to QC flags 1 and 2 only. 13,622 of those are archive levels the verification refuses to score, and it says so

Say casts, not floats. A float drifts and reports repeatedly. Twenty-five marks were made by seventeen instruments; calling the marks floats puts eight instruments in the Bay of Bengal that are not there.

Chlorophyll needs care. We serve it as instrument data from BGC floats (2,120 measured levels, contemporary). We refuse INCOIS's gridded chlorophyll because that series ends in 2020 and would sit on a 2026 time scrubber six years out of step. Do not let the two get merged.

WHERE THE METHODS COME FROM

When a judge asks "where did you get this method from", these are the names. The two to memorise are **Ryan 2015** and **Wong 2020**.

What we did	Where it came from
Model-versus-instrument skill scoring in observation space, per depth band	Ryan et al. 2015, GODAE OceanView Class 4
Keep only QC flags 1 and 2 on every instrument profile	Wong et al. 2020, Argo and its QC flags
Density from temperature, salinity and pressure	TEOS-10, IOC, SCOR and IAPSO 2010
The API shape: bounding box plus depth plus time returns profiles	Tucker et al. 2020, Argovis
3D ocean model fields in a browser on a Cesium globe	Qin et al. 2021
Stop drawing a layer once it cannot change the picture; adaptive sampling	Yu, Qin and Xu 2025, WebGPU ocean volume rendering
Serving through OGC WMS over CF-NetCDF	Blower et al. 2013, ncWMS
Storage as Zarr chunks rather than whole NetCDF files	Signell and Pothina 2019
Extract features on the server, ship light geometry, not whole grids	Liu, Silver and Bemis 2019
The assistant calls pre-registered tools and never writes code	Khanal et al. 2024 (FathomGPT); GeoJSON Agents 2025
The Indian validation precedent	Balakrishnan Nair et al. 2013, INCOIS Ocean State Forecast validation, including cyclone Thane
Formats, layers and alerts	CF Conventions, OGC WMS 1.3.0 and WCS 1.0.0, OASIS CAP v1.2

THE NUMBER, AND THE SENTENCE THAT MUST FOLLOW IT

Figure	Value	Per depth band (RMSE, degC)
Bias (model minus observation)	+0.026 degC	0 to 10 m: 0.32 10 to 50 m: 1.34 50 to 100 m: 2.07 100 to 200 m: 1.35
RMSE (root mean square error)	0.602 degC	200 to 500 m: 0.43 500 to 1000 m: 0.12 1000 to 2000 m: 0.12
Mean absolute error	0.238 degC	
Matched model-observation pairs	11,718	The model is superb in the deep ocean and worst across the thermocline at 50 to 100 m, which is exactly where the gradient is steepest and exactly the layer a cyclone forecaster reads. That is a real, useful, defensible finding.
Casts and instruments contributing	23 casts, 15 instruments	

This sentence must travel with every one of those numbers. The INCOIS analysis assimilates these same Argo profiles, so the residual measures how closely the analysis fits data it has already seen. It bounds the analysis error and is NOT forecast skill. It is also a 10-day averaged field compared against instantaneous profiles, which carries a representativeness error no interpolation can remove. This is printed on our own screen, in red, above our own number. We also publish what we refused to score: 4,672 levels did not become pairs, each counted by reason.

QUESTIONS TO EXPECT, AND THE ANSWERS

- "Doesn't INCOIS already have a 3D portal?"
Yes, and never say otherwise. Their Digital Ocean portal advertises 3D and 4D visualisation, and it is a data-management and fusion portal whose 3D is a globe with draped layers and time animation. We consume their own ERDDAP, so there is no data duplication, and we add what it does not do: true volumetrics, a skill number beside the field, and natural-language control.
- "How is this different from Windy or earth.nullschool?"
Those are surface animations. The ocean state that matters for cyclone intensification, fisheries and search-and-rescue lives below the single pixel they draw. We render the whole column, fuse the instruments into it, and quantify how far the model sits from them.
- "Where does your data come from? Is it real?"
All of it, all open. INCOIS's own ERDDAP for the analysis, the Argo Global Data Assembly Centre at Ifremer for the floats, RAMA for the moored buoy, Copernicus Marine for the currents, NDMA's SACHET for the alerts. Every panel on screen prints its own dataset name and timestamp.
- "How do you know your numbers are right?"
Three answers in this order. The verification card scores us against real instruments and prints the result. 567 automated tests, eleven of which drive a real browser through the whole demo. And the density calculation reproduces the published TEOS-10 check values to six decimal places.
- "Is the AI making things up?"
It cannot. Every figure in every answer has to come from a tool call against real data, and there is code that reads the finished sentence and withholds it if a number appears that no tool produced. Not corrects it, refuses it. And be straightforward: there is no language model in it yet, it is a rule-based router over real tools, and it prints that under every answer.
- "Your glider data is from 2018. Why?"
Because there is no glider in the Bay of Bengal today. We checked three archives; on 1 September the entire Copernicus live feed held one Indian Ocean glider and it was in the Mozambique Channel. The problem statement asks for gliders to be co-displayed, so we show a real Rutgers deployment from 2018, draw a ring around every one of its marks, print the date and the reason when you click it, and make the verification refuse all 13,622 of its levels rather than score an eight-year-old dive against a 2026 field. The alternative was not a newer glider. It was no glider.
- "What is your biggest weakness?"
Answer it straight, having one ready is a strength: the Docker deployment has never been run, because Docker is not installed on the build laptop; and the assistant has no language model in it yet. Both are written down, and neither is hidden in the demo.

THE SINGLE BEST THING TO SHOW, IN ORDER

If they only stay two minutes:

- 1 Drag the depth cursor from 5 m to 2000 m. They see a 3D water column in a browser. Ten seconds.
- 2 Click an Argo float. Its real measured profile appears against the model at the same place and time. Twenty seconds. If they look interested, click the glider next: the line of darts past Sri Lanka is one instrument flying, and the panel says in its own words why a 2018 dive is on a 2026 scene.
- 3 Point at the verification card. "Every ocean viewer draws the model. This one prints how wrong it is: 0.60 degC over 11,718 matched measurements. And underneath, in red, our own caveat saying what that number is not." Thirty seconds. This is the one they will remember.
- 4 Switch the variable to SIG0. "This is density. Nobody measures it; it is computed from temperature and salinity through the international equation of state, by a plugin, which is the extensibility requirement working." Twenty seconds.